

IS SPIKE-DRIVEN SELF-ATTENTION NECESSARY? THE INEFFICIENCY OF JACCARD ATTENTION IN SPIK- ING SENTENCE EMBEDDINGS

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ABSTRACT

Spiking Neural Networks (SNNs) provide an energy-efficient alternative to conventional Deep Neural Networks by utilizing discrete, event-driven temporal spikes. However, adopting Transformer-based paradigms into the spiking domain typically requires mapping complex Self-Attention mechanisms into spike operations. In this paper, we present an in-depth mathematical formulation and empirical ablation of a Spike-Driven Self-Attention layer—modeled around a Jaccard-like spike intersection metric—to evaluate its true utility in semantic sentence embedding tasks. Through comprehensive knowledge distillation, we demonstrate that incorporating Jaccard Attention yields a severe execution bottleneck, quadrupling training time and doubling inference latency, while offering a negligible absolute accuracy gain (+0.0048 Pearson correlation). We propose a purely recurrent, attention-free Spiking Sentence Embedder operating exclusively through Leaky Integrate-and-Fire (LIF) pooling and Backpropagation Through Time (BPTT). Our minimalist architecture incorporates SNN-native hyperpolarization for sequence padding. By explicitly mapping discrete gradients to continuous dense Teacher targets, our model achieves a highly competitive Pearson correlation of 0.8031 on the ALL-STS benchmark. Ultimately, replacing complex spatial attention routing with pure recurrent temporal pooling drastically reduces training and inference time, demonstrating that spatial routing via self-attention provides marginal benefit relative to its computational cost for spiking sentence embedding tasks.

1 INTRODUCTION

The computational overhead of Dense Sentence Embedders (e.g., SBERT, SimCSE) is heavily dominated by the dense matrix multiplications required in standard Scaled Dot-Product Attention (Reimers & Gurevych, 2019; Gao et al., 2021). Spiking Neural Networks (SNNs) alleviate this constraint by operating on sparse, event-driven additions rather than complex multiplications. Recent neuromorphic NLP research has focused on translating Self-Attention to SNNs (Li et al., 2022; Zhu et al., 2023), such as the Sparse Coincidence-Based Semantic Attention (Muhammad, 2026) which relies on spike overlaps. However, a fundamental question persists: *To what extent do the temporal recurrent dynamics of an SNN mitigate the need for explicit $\mathcal{O}(L^2)$ spatial attention mechanisms in sentence embedding tasks?*

In this study, we systematically deconstruct the forward dynamics and learning mechanisms of an SNN sentence embedder down to its core computational primitives. We mathematically define a Jaccard-based Spiking Attention algorithm and contrast it against an attention-free Recurrent Pooler. Our empirical findings across a multi-phase evaluation indicate that the temporal integration capabilities of the recurrent pooler, when optimized via Backpropagation Through Time (BPTT) and Surrogate Gradients, are overwhelmingly sufficient for encoding semantic structures.

2 METHODOLOGY: ARCHITECTURAL DYNAMICS AND BPTT

Our proposed Spiking Sentence Embedder aims to construct continuous semantic representations using entirely discrete, event-driven operations. The pipeline consists of Spike Encoding, a comparative Attention vs. Pooling mechanism, and Knowledge Distillation.

2.1 SPIKE ENCODING AND HETEROGENEOUS DYNAMICS

Raw text is tokenized into vocabulary indices. The Spiking Embedding Layer translates these tokens into temporal spike trains $S_{emb}^{(t)}$ over discrete sequence steps $t \in [1, L]$, where the temporal dimension maps perfectly to sequence length. The membrane potential $V_{emb}^{(t)}$ is modeled via Leaky Integrate-and-Fire (LIF) dynamics:

$$V_{emb}^{(t)} = \min \left(1.0, \beta_{emb} V_{emb}^{(t-1)} + X W_{emb} \right) - S_{emb}^{(t)} \theta_{emb} \quad (1)$$

When the pre-reset potential exceeds the threshold θ_{emb} , a spike $S_{emb}^{(t)} = 1$ is fired. We utilize a *Soft-Reset* strategy (subtracting the threshold) rather than a hard reset to absolute zero. Preserving these fractional sub-threshold residual potentials prevents catastrophic semantic information loss across time. Furthermore, we introduce biological heterogeneity: the leak decay β_{emb} and thresholds θ_{emb} are initialized with unique, uniformly distributed values across dimensions (e.g., $\beta \in [1 - 2^{-2}, 1 - 2^{-7}]$). This forces multi-timescale temporal processing immediately upon sequence ingestion.

Sequence Padding and Hyperpolarization: To handle variable sequence lengths efficiently without relying on explicit attention padding masks, we introduce an SNN-native hyperpolarization technique. The embedding weights associated with the ‘PAD’ token (index 0) are explicitly overridden to -1.0 . Consequently, during padding time-steps, the incoming synaptic dot product yields a strongly negative current, hyperpolarizing the membrane potential ($V < 0$). This guarantees that padding tokens emit exactly zero spikes, completely eliminating noise accumulation during temporal pooling.

2.2 ABLATION BASELINE: SPIKE-DRIVEN JACCARD ATTENTION

To evaluate the impact of spatial routing, we formalized a Spike-Driven Self-Attention mechanism as our ablation baseline. Unlike floating-point dot products, this method computes affinities purely through Boolean spike intersections (Jaccard index numerator). Embedding spikes S_{emb} are projected into Query (Q), Key (K), and Value (V) spikes using independent LIF neurons. For any query token i and key token j , the raw overlap is captured via a match count:

$$M_{i,j}^{(t)} = \sum_{d=1}^{D_{model}} \left(Q_{i,d}^{(t)} \wedge K_{j,d}^{(t)} \right) \quad (2)$$

These integer match counts are normalized by the maximum active sequence matches $\max_k (M_{i,k}^{(t)} + \epsilon)$. To maintain the purely spiking pipeline, these normalized fractional scores are passed into an auxiliary set of LIF neurons to generate attention mask spikes S_{scores} , which finally gate the Value spikes V via accumulation. While this avoids continuous floating-point multiplications, calculating $\mathcal{O}(L^2 \cdot D_{model})$ Boolean intersections at every single time-step, followed by dual-layered LIF dynamics, imposes severe computational bottlenecks and dramatically inflates execution time.

2.3 THE SUPERIOR RECURRENT POOLER (PROPOSED ARCHITECTURE)

We propose that spatial attention is functionally redundant. In our purely recurrent architecture, $S_{emb}^{(t)}$ bypasses attention and directly feeds into a Spiking Dense BPTT Pooler layer. Because the input spike S_{emb} strictly guarantees binary values $\{0, 1\}$, the dense matrix projection is simplified to conditional accumulations:

$$X_{pool}^{(t)} = \sum_{d=1}^{D_{in}} \begin{cases} W_d & \text{if } S_{emb,d}^{(t)} = 1 \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

By bypassing floating-point multiplications entirely, this step offers massive computational sparsity. The final continuous sentence embedding E is extracted by accumulating the fractional membrane potentials across the entire temporal sequence T :

$$E = \sum_{t=1}^T V_{pool}^{(t)} \quad (4)$$

Learning via BPTT and Add-Only Deltas: The true power of this architecture stems from its back-propagation mechanics. Because the Heaviside step function lacks a valid derivative, we employ a Rectangular Surrogate Mask to propagate errors:

$$\sigma'(V) = \begin{cases} 1 & \text{if } |V - \theta| < \text{window_size} \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

During Backpropagation Through Time, the error $\delta^{(t)}$ at sequence step t is explicitly linked to future time steps via the heterogeneous decay β_{pool} :

$$\tilde{\delta}^{(t)} = \left(\delta_{out}^{(t)} + \delta^{(t+1)} \cdot \beta_{pool} \right) \cdot \sigma'(V_{pool}^{(t)}) \quad (6)$$

This explicit temporal unrolling allows the β parameter to act as a natural contextual router. The pooler inherently learns long-term syntactic dependencies simply by preserving or decaying gradients. Furthermore, during weight updating, the gradient accumulation ΔW also utilizes an Add-Only Delta rule:

$$\Delta W_{in,out} = \sum_t \begin{cases} \tilde{\delta}_{out}^{(t)} & \text{if } S_{emb,in}^{(t)} = 1 \\ 0 & \text{otherwise} \end{cases} \quad (7)$$

This establishes a fully multiplication-free learning dynamic in the forward and BPTT routing phases, rendering the spatial $\mathcal{O}(L^2)$ matrix completely unnecessary.

2.4 KNOWLEDGE DISTILLATION AND ERROR ROUTING

To optimize these dynamics, we distill dark knowledge from a pre-trained dense Teacher. We utilize a Mean Squared Error (MSE) objective over the Cosine Similarity scores. The exact error E is given by:

$$E = \text{Score}_{Teacher} - \sum_{d=1}^{D_{model}} (\tilde{E}_{1,d} \cdot \tilde{E}_{2,d}) \quad (8)$$

where $\tilde{E}$ denotes the z-score normalized pooled embeddings. The error gradient is derived symmetrically across the representation space:

$$\frac{\partial \mathcal{L}}{\partial \tilde{E}_{1,d}} = E \cdot \tilde{E}_{2,d} \cdot m \quad (9)$$

$$\frac{\partial \mathcal{L}}{\partial \tilde{E}_{2,d}} = E \cdot \tilde{E}_{1,d} \cdot m \quad (10)$$

This gradient is then injected directly into the temporal unrolling sequence via BPTT. The contrastive margin $m = 0.05$ acts as a dampening factor that restricts explosive updates from sparse binary state flips. This end-to-end framework allows the discrete, event-driven parameter space to safely converge toward the continuous dense Teacher topology.

3 EXPERIMENTAL EVALUATION

The models are implemented in native Rust to guarantee deterministic execution, avoiding PyTorch simulation artifacts. We trained over 20 epochs across 29,720 validation pairs from ALL-STs.

Table 1: Phase 1: Dimensionality Scaling (Recurrent Pooler)

D_{model}	ALL-STS Pearson	Inference Latency (ms/pair)	Train Time (seconds)
32	0.6713	0.718	681
64	0.7091	0.911	1,078
128	0.8038	1.341	2,025
256	0.8079	1.946	4,753

3.1 PHASE 1: DIMENSIONALITY SCALING

Using the Distillation strategy with the Recurrent Pooler, we scaled the embedding dimension D_{model} to analyze representation capacity.

The SNN scales impeccably to 256 dimensions, shattering the 0.80 Pearson correlation barrier, a massive improvement over prior Spike Coincidence Attention baselines (Muhammad, 2026) ($r = 0.758$).

3.2 PHASE 2: ABLATION OF SPIKE-DRIVEN ATTENTION

Fixing the dimension to $D_{model} = 256$, we performed an architectural ablation against the Jaccard Attention and the minimalist Recurrent Pooler.

Table 2: Phase 2: Ablation of Spike-Driven Attention ($D_{model} = 256$)

Architecture Variant	ALL-STS Pearson	Inference (ms/pair)	Train Time (seconds)
No-Attention (Recurrent)	0.8031	0.932	1,165
Jaccard Attention	0.8079	1.946	4,817
Homogeneous-Init (Attn)	0.7996	1.924	4,328

Discussion on Efficiency: The empirical data suggests that Spike-Driven Self-Attention provides marginal benefit relative to its computational cost for this specific sentence embedding context. While Jaccard Attention yields a negligible raw accuracy bump ($\Delta = +0.0048$), it introduces a severe computational bottleneck. Incorporating the explicit $L \times L$ attention matrix effectively quadruples the total training time (1,165 seconds vs 4,817 seconds) and doubles the execution latency (0.932 ms vs 1.946 ms) per sentence pair.

The Recurrent Pooler achieves computational superiority by bypassing spatial routing entirely. The BPTT gradients of the Recurrent Pooler’s temporal context ($\beta V_{pool}^{(t-1)}$) prove highly expressive, highlighting recurrent SNNs as a highly competitive and efficient topology for sequences under constrained budgets.

4 LIMITATIONS

While our attention-free recurrent SNN achieves remarkable sparsity, the peak correlation (0.8031) remains marginally below the theoretical limit of the full-precision Teacher model (> 0.85), indicating an inherent compression bottleneck in discrete binary signaling. Furthermore, our empirical sequence length is capped at $L = 128$; verifying temporal decay limits over much longer contexts (e.g., document-level embeddings) requires future investigation.

5 CONCLUSION

Through detailed mathematical formulation and rigorous low-level structural ablation, we demonstrate that forcing Spatial Attention onto Spiking Neural Networks produces an unjustifiable computational bottleneck. The biological temporal decay properties inherent to LIF neurons, when trained

via Surrogate Gradients and BPTT, naturally encapsulate complex sentence semantics without requiring quadratic sequence routing. We conclude that Spike-Driven Self-Attention is redundant for dense semantic representations, paving the way for ultra-efficient, purely recurrent SNN embedders.

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